Engineering Geology Ltd
CONSULTING GEOTECHNICAL, GEOLOGICAL AND EARTHQUAKE ENGINEERS

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Ref: 8447

Ray and Desma Nicklin
PO Box 404243
PUHOI 0951

14 September 2017

Dear Ray and Desma,

RE: PROPOSED SUBDIVISION
53B REMIGER ROAD, PUHOI
Geotechnical Investigation

1.0 INTRODUCTION

As requested, we have carried out a Geotechnical Investigation for a single lot within a proposed 5-lot subdivision located at 53b Remiger Road. A sketch plan of the proposed subdivision prepared by Cato Bolam Consultants showing the location of the proposed building platform and boundaries was provided to us and is appended.

The investigation has been carried out to assess the subsoil conditions on site, assess the stability of the site and provide geotechnical recommendations for the proposed development. We have also provided comments on the ground condition for a proposed right of way (r.o.w). It is understood that this report will be used to support a Resource Consent application to Auckland Council.

2.0 SITE DESCRIPTION AND PROPOSED DEVELOPMENT

The property is located on the southern side of Remiger Road, Puhoi. The property is legally described at Lot 8 DP 208981 and covers an area of approximately 574262m². The property currently consists of a working farm with a series of farm buildings and equipment located throughout. Current access to and around the farm is via a series of gravel farm tracks which extend from a right of way located in the north-eastern corner. The topography of the farm generally comprises a low-lying area to the north which transitions into a series of steeply sloping ridges and valleys over the remainder of the property.

The subject lot is the southernmost lot of the proposed 5-lot subdivision. The lot is situated within the north-eastern portion of the farm, to the south-west of an existing farm house/shed. As shown on the attached Drawing 8447-1, it is proposed for a building platform to be located on the knoll of a narrow northwest trending ridge within the southern portion of the lot. The side slopes of this ridge generally fall to a series of watercourses located along the inverts of two valleys to the east and west at a gradient of between 1V:1H and 1V:5H. The area currently comprises grazed grass fields with several mature trees scattered through the lot.

From our site walkover, the western and southern slopes of the ridge show signs of past slope movement with several head scarps present near the crest of the ridge and evidence of shallow soil creep.

Directors: Trevor Matuschka BE(Hons), PhD, FIPENZ, CPEng, IntPE, Cat. A Rec. Eng Jeremy Yeats BSc(Civ. Eng), DIC, MSc, MIPENZ, MICE, CPEng, IntPE, Cat. A Rec. Eng
Rambod Amigh BSc, MEStud, MIPENZ, MIEAust, CPEng, IntPE, RPEQ Consultant: Christopher P. Gulliver BSc, BE(Hons), MIPENZ, CPEng, IntPE, PEngGeol

The R.O.W is proposed through the centre of the subdivision as shown on the attached Cato Bolam Sketch. The R.O.W will provide access to the proposed lots, including the subject lot. The southern portion of the R.O.W will run along an existing gravel track, while the northern portion will be over an existing grass paddock field.

3.0 FIELDWORK

The fieldwork was carried out on 31st August 2017 and comprised a walkover survey by a Senior Geotechnical Engineer and the drilling of 6 hand auger boreholes designated BH1 to BH6. BH1, BH2, BH3 and BH5 were drilled to 4.8m depth. BH4 and BH6 were drilled to 4.6m and 3.6m depth were hard ground was encountered.

Pilcon hand shear vane tests were carried out at approximately 0.3m depth intervals within the boreholes to measure the in-situ shear vane strengths of the subsoils. In addition, the boreholes were probed with a Scala Penetrometer to depths of 6.9m for BH1 and BH2, 5.9m for BH3, 4.7m for BH4, 6.9m for BH5 and 4.8m for BH6. The soil was logged and described in general accordance with the New Zealand Geotechnical Society's "Field Description of Soil and Rock" 2005.

The approximate location of the boreholes together with existing on-site features and boundaries are shown on the attached drawing number 8447-1. Descriptions of the soils encountered in the boreholes together with the measured shear vane strengths and Scala Penetrometer test results are presented on the attached log sheets.

4.0 GROUND CONDITIONS

The Institute of Geological and Nuclear Sciences geological map "Auckland Urban Area", Sheet R11 (1992) to a scale of 1:50,000 indicates that the site is underlain by mudstone and siltstone of Pakiri Formation within the Miocene aged Waitemata Group. The subsoils encountered on site comprised residual Waitemata Group deposits, consistent with the geologic map.

Topsoil was present across site to between 0.1m and 0.3m depth. Colluvium was encountered in BH3 to 1.2m depth. The colluvium comprised clayey silt and silty clay of low to moderate plasticity. Measured shear vane strengths ranged between 84kPa to 154kPa.

Below the topsoil and colluvium, the boreholes encountered residual soil to the base of the boreholes. The residual soil comprised clay and silt of low to high plasticity. Measured shear vane strengths were typically greater than 80kPa and ranged to more than 200kPa. Weaker readings were encountered in BH4 between 1.8m and 3.3m depth (42kPa to 56kPa), in BH1 at 1.8m (70kPa) and in BH6 at 1.2m(74kPa).

Scala blow counts in BH3, BH4 and BH6 exceeded 20 blows per 100mm (Practical refusal) at 5.9m, 4.7m and 4.9m respectively. Blow counts in BH1, BH2 and BH5 did not exceeded 20 blows per 100mm down to 6.9m (limit of equipment).

Based on visual characteristics, we assess the near surface soils as likely to be reactive, i.e. susceptible to seasonal soil shrinkage and swelling (Class M in terms of AS2870:2011).

5.0 GROUNDWATER

Groundwater seepage was encountered during drilling in BH4 at 2.0m depth. The remaining boreholes remained dry. The boreholes were drilled within the winter season following a period of heavy rain. It is likely that the groundwater levels will naturally fluctuate and the measured groundwater table represents the period of higher winter water levels.

6.0 STABILITY ANALYSIS

We have carried out slope stability analyses along Cross Section A-A' (refer to attached Drawing No. 8446-2) using the SLOPE/W computer program. The section was prepared using measurements taken on site. We have carried out analyses on both the north-eastern and southwestern profiles of the ridge. The results of the analyses are shown on the attached Figures 1 to 5. We have assumed conservative effective shear strength parameters for the soil layers in accordance with Auckland Council's generally accepted guidelines for the evaluation of slope stability.

The analyses were carried out using two groundwater conditions referred to as "moderately conservative" and "worst credible". The moderately conservative groundwater condition was taken at around 1.0-2.0m below the surface (refer to Figures 1 and 4). The worst credible groundwater level was taken at around 0.5-1.0m below the surface (refer to Figures 2, 3 and 5).

The results of the analyses show that the Factor of Safety (FoS) for the southwestern profile is about 1.46 under moderately conservative conditions and 1.24 for the worst credible ground water conditions. The FoS for both these conditions is below the minimum values considered acceptable (1.5 for moderately conservative and 1.3 for worst credible). As a result, we reran the analysis under worst credible conditions with the addition of a Building Restriction Line (BRL) setback around 10m from the Top of Bank (T.o.B). With the addition of the BRL, the FoS increased to 1.35, above the minimum value of 1.3.

The results of the analyses for the north-eastern profile show the FoS is about 1.9 under moderately conservative conditions and 1.4 for the worst credible ground water conditions. Both these values are above the minimum value considered acceptable (as shown above). It is therefore considered that the property east of the BRL is suitably stable for development subject to the recommendations provided in this report.

7.0 DISCUSSION

Based on our observations on site, boreholes and the results of the stability analyses of the southwestern profile of the ridge, a Building Restriction line is proposed, as shown on the attached Drawing 8447-1. This BRL is setback a minimum of 10m from the Top of Bank (T.o.B) along the western and southern side of the ridge. This positions the building platform on the eastern slopes of the ridge, away from the existing head scarp.

Due to the building platform being on/in proximity to sloping ground, there is a risk of ongoing soil creep. Soil creep is the lateral migration of the soil due to seasonal shrink/swell and gravity pulling the soil downslope. As a result, it is considered that all foundations should be fully pile supported. Piles should be founded below any fill in very stiff residual soil a minimum of 5.0m below existing ground level and designed to resist a lateral earth pressure from soil creep over the upper 1.5m of the pile.

In addition, it is recommended that a level building platform is created by cutting down the crest of the ridge. This will act to create a level platform and also reduce the surcharge on the slope below. No fill should be places on site during the construction of the platform and all earthwork should comprise predominantly cut earthworks. The final position of the building platform will be confirmed following the review of the earthworks plan.

The majority of the proposed R.o.W will be located along an existing gravel track. The northern section will run through an existing grass paddock and will likely require minor earthworks (i.e. cuts less than 1.5m).

Once the earthworks and foundations plans have been prepared, these should be submitted to be reviewed by Engineering Geology Ltd to confirm that our recommendations have been carried out and assess the need for any further investigation.

8.0 DEVELOPMENT RECOMMENDATIONS

Our development recommendations are as follows.

1) Prior to finalising the design, the foundation and earthworks plans for the dwelling must be reviewed and approved by a Geotechnical Engineer to ensure that our recommendations have been correctly interpreted. Additional site-specific investigations during the building consent stage may be necessary depending on the proposed development.

2) All foundations, floor slabs and services into any dwelling on the building sites must be designed for class M soils in terms of AS2870:2011 unless laboratory testing is carried out to confirm the expansiveness of the soil.

3) Earthworks should predominantly comprise cut with no filling to prevent surcharging on the slope. Spoil from excavations may be placed on site provided it is not placed adjacent to or over any steep slope, subject to the advice and approval of a Geotechnical Engineer. All cuts greater than 1m depth should be retained by Engineer designed retaining walls or battered back to a slope of no steeper than 1V:3H, subject to the advice and approval of a Geotechnical Engineer.

4) The building platform should be kept to the east of the BRL as shown on Drawing 8447-1. Building foundations should be fully pile supported. Piles should be founded below any fill a minimum of 5.0m below final ground level or 7 pile diameters, whichever is greater. Piled foundations may be designed using Ultimate Limit State for a Dependable End Bearing of 450kPa (900kPa Ultimate) and Dependable Skin Friction of 15kPa (30kPa Ultimate). Skin friction to 0.6m depth should be ignored. The piles should be founded in very stiff residual soil (Cu>100kPa), subject to inspection and confirmation by a Geotechnical Engineer or Engineering Geologist. In addition, the piles should be designed to resist soil creep pressure over the top 1.5m of the natural ground, assuming the lateral pressure acts over a width of 3 pile hole diameters and using a lateral earth pressure coefficient of k = 0.7.

5) If any building is carried out to west of the BRL this would require the construction of an in-ground retaining wall on the downslope side of any dwelling. The in-ground pile wall should be subject to specific design in consultation with a Geotechnical Engineer to provide specific design parameters. This may also require the need to drill machine boreholes to assess ground conditions over the length of the piles.

6) Floor slabs may be cast on grade on a layer of adequately compacted hardfill provided all topsoil is removed below the slab. All floor slabs should be designed according to recommendation 2 above. Only good quality clean hardfill should be used as basecourse below floor slabs. Any hardfill placed must be compacted to an acceptable engineering standard.

7) For retaining walls, the following design parameters may be adopted. Any retaining wall on sloping ground that is not part of the building structure must comprise an engineer designed timber pole wall unless otherwise agreed by a Geotechnical Engineer.

Effective friction angle	= 30°
Effective cohesion	= 0
Unit weight of soil	= 18kN/m^3
Undrained shear strength (Ultimate)	= 60 kPa

A strength reduction factor of 0.5 should be applied to the undrained shear strength, The walls must be designed for any surcharge loads and taking into account the slope of the land below the walls which will increase the embedment depths using a 'Broms' type of analysis.

8) Any driveway may be designed assuming a CBR of 3 provided topsoil and any weak organic soils and unsuitable fill are removed prior to placing basecourse and subject to inspection and approval of the sub-grade following stripping.

9) The subgrade surface under any foundation, floor slab and the driveway should not be allowed to dry out or be remoulded by construction work and should be protected with a layer of base course, or similar, immediately following excavation and trimming to the design profile. Any cut to form a level platform prior to retaining must be covered with polythene to prevent the soil from drying out. Floor slabs should not be poured on a subgrade which has been allowed to dry out and desiccate. Should cracking of any clay subgrade of more than 10mm be observed, the advice of a Geotechnical Engineer should be obtained before placing hardfill or pouring slabs.

10) The drive and building platform should be shaped to shed storm water evenly with no concentrated flows. Water tank overflow and any concentrated flows of storm water from the drive or other hard surfaces should be piped or channelled to the base of the property in a manner not to cause erosion. Any overland flow should be channelled away from the building site.

11) A proposed location for a wastewater field (including indicative size – subject to detailed design) is shown on the attached Drawing 8447-1. The final design of the disposal field must be carried out in accordance with TP 58 for the proposed development. The surfical soils on site generally consist of up to 0.3m of organic silt (topsoil) underlain by silty clay and clayey silt to depths exceeding 1.5m. We assess the topsoil to be soil Category 4 and the underlying soils to be Category 6 in terms of the definitions defined in TP 58. The effluent disposal system should be designed on this basis.

12) The soils on the site are considered to be moderately susceptible to seasonal shrinkage and swelling. The effects of which can be exacerbated by trees (especially exotic varieties), hedges and plants having a high-water demand, which, should not be planted near the building as they can withdraw moisture from the soil and cause any shallow foundations or ground bearing floor slabs to settle. As a guide to class M soils trees should be planted no closer to the building than 0.75 times the mature height of the tree.

13) During earthworks and foundation construction the site should be examined by a Geotechnical Engineer or Engineering Geologist to measure soil strengths within foundation excavations, as well as to provide advice on any other geotechnical issues which may become apparent during construction.

Where Engineering Geology Ltd is required to carry out site inspections during construction and provide certification or a Producer Statement – Construction Review, it is requested that we be supplied a copy of the Building Consent Conditions. We request that a minimum notification of 24 hours be given for any site inspections and note that we will be unable to issue a Producer Statement without site inspections at the appropriate stages during construction.

We note that site inspections during construction and associated certification are not included within the budget for the preparation of the geotechnical report and will be subject to additional costs

9.0 LIMITATIONS

Recommendations and opinions in this report are based on 6 Hand Auger boreholes. While the nature and continuity of the subsoil conditions away from the boreholes is inferred, it is possible that the actual conditions could vary from those assumed. Should variations in the subsoil conditions described in this report be found to exist, or if the building proposal is altered significantly from that shown on the drawings supplied to us, then it is essential that Engineering Geology Ltd be contacted as it may affect the design parameters recommended above and additional investigation may be required.

This report has been prepared solely for the benefit of Ray and Desma Nicklin as our client with respect to the brief to assess the subsoil conditions on site and provide geotechnical recommendations for the proposed development on the property and Engineering Geology Ltd accepts no liability to any other party in relation to this report. The reliance by other parties on the information or opinions contained in this report shall, without our prior review and agreement in writing be at such parties' sole risk.

We would be happy to provide any further advice that you may require.

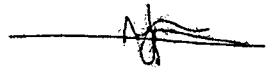
Yours faithfully
ENGINEERING GEOLOGY LTD

Report prepared by



Joseph Wilson (Engineering Geologist)

Reviewed by



Rambod Amigh (CPEng)

Encl
Sketch Plan by Cato Bolam
Borehole logs BH1-BH6
Scala Penetrometer Logs BH1-BH6
Borehole location plan – Dwg No. 8447-1.
Cross Section A-A' – Dwg No. 8447-2
Stability Figures 1 - 5

Address hazard r



"Plan prepared by Cato Bolam Consultants"

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HAND AUGER LOG

BOREHOLE No.: BH1

SHEET 1 OF 1

Job No.: 8447

PROJECT: Remiger 53b

LOCATION: 53b Remiger Road, PUHOI

COORDINATES: **RL GROUND:** **DATE:** 31-08-2017

GRID: **DATUM:** **HOLE DEPTH:** 4.8m

GEOLOGICAL UNIT	SOIL MATERIAL DESCRIPTION	DEPTH / RL	GRAPHIC LOG	DEPTH (m)	MOISTURE CONDITION	CONSISTENCY / DENSITY	SAMPLES	WATER CONTENT (%)	WATER LEVEL	CORRECTED VANE SHEAR STRENGTH (kPa) ● Field Vane (BS 1377) ○ Remoulded Field Vane (50 100 150)	FIELD TESTS
TS	Organic SILT; dark brown	0.0 / 0.1	TS								
Waitemata Group	Silty CLAY, trace sand; light brown. Very stiff, moist, high plasticity.	-0.1								○ 137 ● 45	SV: 0.3m, 137 / 45 kPa (3.1)
	orange with light brown	0.5 / -0.5								○ 154 ● 53	SV: 0.6m, 154 / 53 kPa (2.9)
					M					○ 98 ● 42	SV: 0.9m, 98 / 42 kPa (2.3)
	with trace highly weathered gravel(fine-medium)	1.0 / -1.0		1		VSt				● 200+	SV: 1.2m, 200+ kPa
										○ 116 ● 53	SV: 1.5m, 116 / 53 kPa (2.2)
	no gravel, some fine sand; pinkish red	1.6 / -1.6									
	stiff, moist-wet	1.8 / -1.8				St				○ 70 ● 28	SV: 1.8m, 70 / 28 kPa (2.5)
	Clayey SILT, some sand; reddish pink. Very stiff, moist-wet, low-moderate plasticity.	2.0 / -2.0		2						● 200+	SV: 2.1m, 200+ kPa
	pink, white and black	2.4 / -2.4				VSt			Groundwater Not Encountered	○ 140 ● 39	SV: 2.4m, 140 / 39 kPa (3.6)
										○ 165 ● 34	SV: 2.7m, 165 / 34 kPa (4.9)
	Silty CLAY, trace fine sand; pink, white and red with black specks. Very stiff, moist-wet, moderate plasticity.	3.0 / -3.0		3						○ 109 ● 38	SV: 3.0m, 109 / 38 kPa (2.9)
	hard	3.3 / -3.3			M-W	H				● 200+	SV: 3.3m, 200+ kPa
										● UTP	SV: 3.6m, UTP
	very stiff	3.9 / -3.9		4		VSt				○ 148 ● 34	SV: 3.9m, 148 / 34 kPa (4.4)
										○ 112 ● 34	SV: 4.2m, 112 / 34 kPa (3.3)
										○ 154 ● 70	SV: 4.5m, 154 / 70 kPa (2.2)
	hard	4.7 / -4.7 / 4.8				H				● UTP	SV: 4.8m, UTP

EOH: 4.80 m

NOTES:

1. E.O.B@4.8m (Target Depth).
2. For ground conditions below 4.8m, refer to Scala Penetrometer Log.

LOGGED: DB

CHECKED: RA

DRILL TYPE: 50mm Hand Auger

EGL - Hand Auger - Test Pit v7 - 13-Sep-17 11:11:10 AM - Produced with Core-GS by Geroc

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HAND AUGER LOG

BOREHOLE No.: BH2

SHEET 1 OF 1

Job No.: 8447

PROJECT: Remiger 53b

LOCATION: 53b Remiger Road, PUHOI

COORDINATES: **RL GROUND:** **DATE:** 31-08-2017

GRID: **DATUM:** **HOLE DEPTH:** 4.8m

GEOLOGICAL UNIT	SOIL MATERIAL DESCRIPTION	DEPTH / RL	GRAPHIC LOG	DEPTH (m)	MOISTURE CONDITION	CONSISTENCY / DENSITY	SAMPLES	WATER CONTENT (%)	WATER LEVEL	CORRECTED VANE SHEAR STRENGTH (kPa) ●Field Vane (BS 1377) ○Remoulded Field Vane 50 100 150	FIELD TESTS
TS	Organic SILT; dark brown.	0.0	TS TS								
		0.3									SV: 0.3m, 148 / - kPa (N/A)
Waitemata Group	Silty CLAY; orange. Very stiff, moist, moderate plasticity.	-0.3									
	orange and light brown	0.6 / -0.6									SV: 0.6m, 175 / 74 kPa (2.4)
		0.9									SV: 0.9m, 162 / - kPa (N/A)
	Clayey SILT; streaked orange, light pink and light brown. Very stiff, moist, moderate plasticity.	-0.9		1		VSt					SV: 1.2m, 135 / 67 kPa (2.0)
	reddish brown and pinkish grey	1.5 / -1.5									SV: 1.5m, 111 / - kPa (N/A)
											SV: 1.8m, 108 / 54 kPa (2.0)
	reddish brown, light pink and orange	2.1 / -2.1		2					Groundwater Not Encountered		SV: 2.1m, 108 / - kPa (N/A)
	stiff	2.4 / -2.4			M						SV: 2.4m, 92 / 40 kPa (2.3)
											SV: 2.7m, 84 / - kPa (N/A)
				3		St					SV: 3.0m, 88 / - kPa (N/A)
											SV: 3.3m, 92 / 54 kPa (1.7)
	very stiff	3.6 / -3.6									SV: 3.6m, 121 / - kPa (N/A)
				4		VSt					SV: 3.9m, 146 / - kPa (N/A)
											SV: 4.2m, 148 / 81 kPa (1.8)
											SV: 4.5m, 162 / - kPa (N/A)
		4.8									SV: 4.8m, 175 / - kPa (N/A)

EOH: 4.80 m

NOTES:

1. E.O.B@4.8m (Target Depth).
2. For ground conditions below 4.8m, refer to Scala Penetrometer Log.

LOGGED: JWil

CHECKED: RA

DRILL TYPE: 50mm Hand Auger

EGL - Hand Auger - Test Pit v7 - 13-Sep-17 11:11:10 AM - Produced with Core-GS by Geroc

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HAND AUGER LOG

BOREHOLE No.: BH3

SHEET 1 OF 1

Job No.: 8447

PROJECT: Remiger 53b

LOCATION: 53b Remiger Road, PUHOI

COORDINATES:

RL GROUND:

DATE: 31-08-2017

GRID:

DATUM:

HOLE DEPTH: 4.8m

GEOLOGICAL UNIT	SOIL MATERIAL DESCRIPTION	DEPTH / RL	DEPTH (m)	MOISTURE CONDITION	CONSISTENCY / DENSITY	SAMPLES	WATER CONTENT (%)	WATER LEVEL	CORRECTED VANE SHEAR STRENGTH (kPa) ●Field Vane (BS 1377) ○Remoulded Field Vane	FIELD TESTS
TS	Organic SILT; dark brown	0.0 / 0.1								
Colluvium	Clayey SILT; orange. Stiff, dry-moist, low-moderate plasticity.	-0.1 / 0.3		D-M						
	Silty CLAY, trace sand and fine-coarse gravel; orange and red. Stiff, moist, moderate plasticity.	-0.3 / 0.6			St					SV: 0.3m, 84 / 14 kPa (6.0)
	very stiff	-0.6 / 1.0	1							SV: 0.6m, 154 / 42 kPa (3.7)
	Clayey SILT, minor sand and trace fine-medium gravel; pink orange. Very stiff, moist, moderate plasticity.	-1.0 / 1.2			VSt					SV: 0.9m, 126 / 17 kPa (7.5)
Waitemata Group	Silty CLAY, minor sand; pink and light grey. Very stiff, moist, moderate plasticity.	-1.2 / 1.5								SV: 1.2m, 101 / 28 kPa (3.6)
	hard	-1.5 / 1.8								SV: 1.5m, 200+ kPa
	limonite staining	-1.8 / 2.1	2		H					SV: 1.8m, 200+ kPa
	light grey with orange staining, very stiff	-2.1 / 2.3								SV: 2.1m, 112 / 28 kPa (4.0)
	orange pink and light grey	-2.3 / 3.2		M				Groundwater Not Encountered		SV: 2.4m, 105 / 39 kPa (2.7)
	orange, light grey and black(MnO)	-3.2 / 4.3	3		VSt					SV: 2.7m, 157 / 56 kPa (2.8)
	hard	-4.3 / 4.8	4		H					SV: 3.0m, 126 / 36 kPa (3.5)
										SV: 3.3m, 95 / 34 kPa (2.8)
										SV: 3.6m, 140 / 70 kPa (2.0)
										SV: 3.9m, 104 / 25 kPa (4.1)
										SV: 4.2m, 157 / 56 kPa (2.8)
										SV: 4.5m, UTP
										SV: 4.8m, UTP

EOH: 4.80 m

NOTES:

1. E.O.B@4.8m (Target Depth).
2. For ground conditions below 4.8m, refer to Scala Penetrometer Log.

LOGGED: DB

CHECKED: RA

DRILL TYPE: 50mm Hand Auger

EGL - Hand Auger - Test Pit v7 - 13-Sep-17 11:11:10 AM - Produced with Core-GS by Geroc

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HAND AUGER LOG

BOREHOLE No.: BH4

SHEET 1 OF 1

Job No.: 8447

PROJECT: Remiger 53b

LOCATION: 53b Remiger Road, PUHOI

COORDINATES: **RL GROUND:** **DATE:** 31-08-2017

GRID: **DATUM:** **HOLE DEPTH:** 4.6m

GEOLOGICAL UNIT	SOIL MATERIAL DESCRIPTION	DEPTH / RL	GRAPHIC LOG	DEPTH (m)	MOISTURE CONDITION	CONSISTENCY / DENSITY	SAMPLES	WATER CONTENT (%)	WATER LEVEL	CORRECTED VANE SHEAR STRENGTH (kPa) ●Field Vane (BS 1377) ○Remoulded Field Vane 50 100 150	FIELD TESTS
TS	Organic SILT; dark brown	0.0 / 0.1	TS								
Waitemata Group	Silty CLAY, trace sand; light brown. Very stiff, moist, high plasticity.	-0.1									SV: 0.3m, 137 / 49 kPa (2.8)
	orange with light brown streaks	0.4 / -0.4									SV: 0.6m, 120 / 48 kPa (2.5)
	some silt and trace fine sand; orange and light grey	0.7 / -0.7		1	M	VSt					SV: 0.9m, 126 / 56 kPa (2.3)
	wet	1.6 / -1.6									SV: 1.2m, 119 / 62 kPa (1.9)
	stiff	1.8 / -1.8			W						SV: 1.5m, 119 / 49 kPa (2.4)
	saturated	2.0 / -2.0		2		St			2.00m, 31-08-2017		SV: 1.8m, 56 / 22 kPa (2.5)
	firm	2.7 / -2.7									SV: 2.1m, 81 / 42 kPa (1.9)
	silty, trace sand; dark grey with orange staining. Very stiff, saturated, high plasticity.	3.3 / -3.3		3	S	F					SV: 2.4m, 53 / 25 kPa (2.1)
											SV: 2.7m, 42 / 22 kPa (1.9)
											SV: 3.0m, 48 / 22 kPa (2.1)
				4		VSt					SV: 3.3m, 98 / 42 kPa (2.3)
	hard	4.2 / -4.2				H					SV: 3.6m, 126 / 42 kPa (3.0)
											SV: 3.9m, 104 / 28 kPa (3.7)
											SV: 4.2m, 200 / - kPa (N/A)
		4.6									SV: 4.6m, 200 / - kPa (N/A)

EOH: 4.60 m

NOTES:

1. E.O.B@4.6m (to hard to auger).
2. For ground conditions below 4.6m, refer to Scala Penetrometer Log.

LOGGED: DB

CHECKED: RA

DRILL TYPE: 50mm Hand Auger

EGL - Hand Auger - Test Pit v7 - 13-Sep-17 11:11:10 AM - Produced with Core-GS by Geroc

ENGINEERING GEOLOGY LTD

HAND AUGER LOG

BOREHOLE No.: BH5

SHEET 1 OF 1

Job No.: 8447

PROJECT: Remiger 53b

LOCATION: 53b Remiger Road, PUHOI

COORDINATES: **RL GROUND:** **DATE:** 31-08-2017

GRID: **DATUM:** **HOLE DEPTH:** 4.8m

GEOLOGICAL UNIT	SOIL MATERIAL DESCRIPTION	DEPTH / RL	GRAPHIC LOG	DEPTH (m)	MOISTURE CONDITION	CONSISTENCY / DENSITY	SAMPLES	WATER CONTENT (%)	WATER LEVEL	CORRECTED VANE SHEAR STRENGTH (kPa) ● Field Vane (BS 1377) ○ Remoulded Field Vane (50 100 150)	FIELD TESTS
TS	Organic SILT; dark brown	0.0 0.2	TS								
Waitemata Group	Silty CLAY; orange and light grey. Very stiff, moist, moderate plasticity.	-0.2		1	M	VSt			Groundwater Not Encountered	● 0.3m: 175 kPa ○ 0.6m: 159 kPa ● 0.6m: 67 kPa ● 0.9m: 175 kPa ○ 1.2m: 148 kPa ● 1.2m: 108 kPa ● 1.5m: 135 kPa ○ 1.8m: 132 kPa ● 1.8m: 61 kPa	SV: 0.3m, 175 / - kPa (N/A) SV: 0.6m, 159 / 67 kPa (2.4) SV: 0.9m, 175 / - kPa (N/A) SV: 1.2m, 148 / 108 kPa (1.4) SV: 1.5m, 135 / - kPa (N/A) SV: 1.8m, 132 / 61 kPa (2.2)
		2.7		2						● 2.1m: 108 kPa ● 2.4m: 108 kPa ● 2.7m: 105 kPa	SV: 2.1m, 108 / - kPa (N/A) SV: 2.4m, 108 / - kPa (N/A) SV: 2.7m, 105 / - kPa (N/A)
	Clayey SILT; light grey. Very stiff, wet, moderate plasticity.	-2.7		3	W					● 3.0m: 108 kPa	SV: 3.0m, 108 / - kPa (N/A)
	light grey and orange	3.0 -3.0									
	dry-moist	3.3 -3.3								○ 3.3m: 119 kPa ● 3.3m: 54 kPa	SV: 3.3m, 119 / 54 kPa (2.2)
	stiff	3.6 -3.6			D-M	St				● 3.6m: 94 kPa	SV: 3.6m, 94 / - kPa (N/A)
	very stiff, moist	3.9 -3.9		4	M	VSt				● 3.9m: 148 kPa ● 4.2m: 148 kPa ● 4.5m: 162 kPa ● 4.8m: 151 kPa	SV: 3.9m, 148 / - kPa (N/A) SV: 4.2m, 148 / - kPa (N/A) SV: 4.5m, 162 / - kPa (N/A) SV: 4.8m, 151 / - kPa (N/A)
		4.8									

EOH: 4.80 m

NOTES:
1. E.O.B@4.8m (Target Depth).
2. For ground conditions below 4.8m, refer to Scala Penetrometer Log.

LOGGED: JWil

CHECKED: RA

DRILL TYPE: 50mm Hand Auger

EGL - Hand Auger - Test Pit v7 - 13-Sep-17 11:11:10 AM - Produced with Core-GS by Geroc

ENGINEERING GEOLOGY LTD

HAND AUGER LOG

BOREHOLE No.: BH6

SHEET 1 OF 1

Job No.: 8447

PROJECT: Remiger 53b

LOCATION: 53b Remiger Road, PUHOI

COORDINATES: **RL GROUND:** **DATE:** 31-08-2017

GRID: **DATUM:** **HOLE DEPTH:** 3.3m

GEOLOGICAL UNIT	SOIL MATERIAL DESCRIPTION	DEPTH / RL	GRAPHIC LOG	DEPTH (m)	MOISTURE CONDITION	CONSISTENCY / DENSITY	SAMPLES	WATER CONTENT (%)	WATER LEVEL	CORRECTED VANE SHEAR STRENGTH (kPa) ● Field Vane (BS 1377) ○ Remoulded Field Vane (50 100 150)	FIELD TESTS
TS	Organic SILT; dark brown	0.0 / 0.2	TS								
Waitemata Group	Silty CLAY; orange. Very stiff, moist, moderate plasticity.	-0.2 / 0.8			M	VSt				SV: 0.3m, 162 / - kPa (N/A)	
	Clayey SILT, some sand; reddish orange and light grey. Very stiff, moist-wet, moderate plasticity.	-0.8 / 1.2		1	M-W					SV: 0.6m, 111 / - kPa (N/A)	
	stiff	-1.2 / 1.5				St				SV: 0.9m, 108 / - kPa (N/A)	
	orange and light grey with grey streaks/specks. dry-moist	-1.5 / 1.8								SV: 1.2m, 74 / - kPa (N/A)	
	very stiff	-1.8 / 2.0								SV: 1.5m, 94 / - kPa (N/A)	
	some black specks	-2.0 / 2.7		2	D-M	VSt			Groundwater Not Encountered	SV: 1.8m, 148 / - kPa (N/A)	
	some clay; reddish brown with some grey and dark grey streaks, hard, moist, low-moderate plasticity.	-2.7 / 3.2		3	M	H				SV: 2.1m, 148 / - kPa (N/A)	
	reddish brown, dry	-3.2 / 3.3			D					SV: 2.4m, 175 / - kPa (N/A)	
	EOH: 3.30 m									SV: 2.7m, 200+ kPa	
										SV: 3.0m, UTP	
										SV: 3.3m, UTP	

NOTES:

1. E.O.B@3.3m (to hard to auger).
2. For ground conditions below 3.3m, refer to Scala Penetrometer Log.

LOGGED: JWil

CHECKED: RA

DRILL TYPE: 50mm Hand Auger

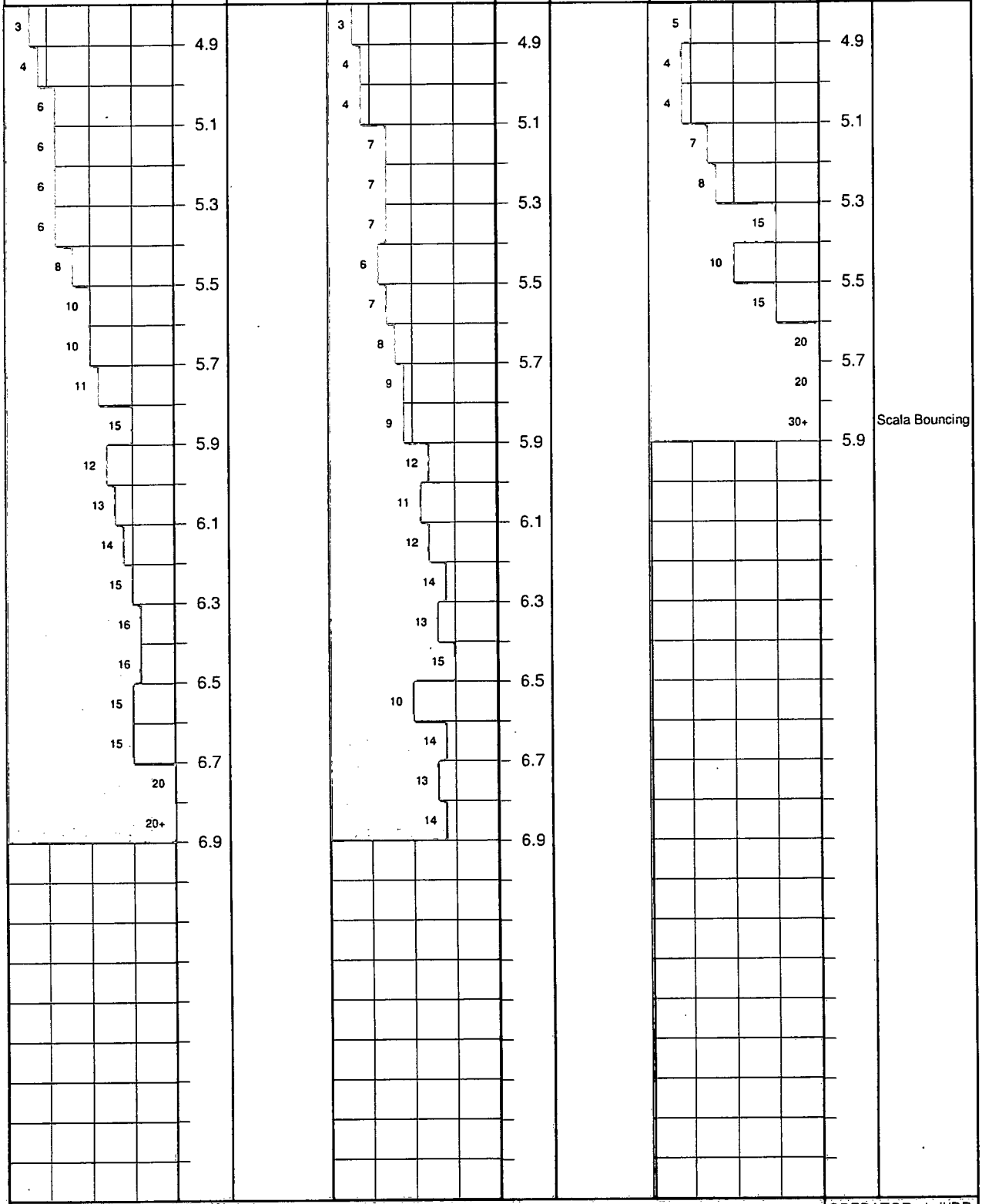
EGL - Hand Auger - Test Pit v7 - 13-Sep-17 11:11:10 AM - Produced with Core-GS by Geroc

Engineering Geology Ltd
CONSULTING GEOTECHNICAL, GEOLOGICAL AND EARTHQUAKE ENGINEERS

SCALA PENETROMETER LOG

REF: 8447 | Sheet 1 of 2

Site:53b Remiger Road, PUHOI



NOTES:

OPERATOR: Jwil/DB

DATE:31/8/2017

Engineering Geology Ltd
CONSULTING GEOTECHNICAL, GEOLOGICAL AND EARTHQUAKE ENGINEERS

SCALA PENETROMETER LOG

REF: 8447 | Sheet 2 of 2

Site:53b Remiger Road, PUHOI

BH 4

Blows / 100 mm	Depth (m)	Interpret. / Comments
30+	4.7	Scala Bouncing

BH 5

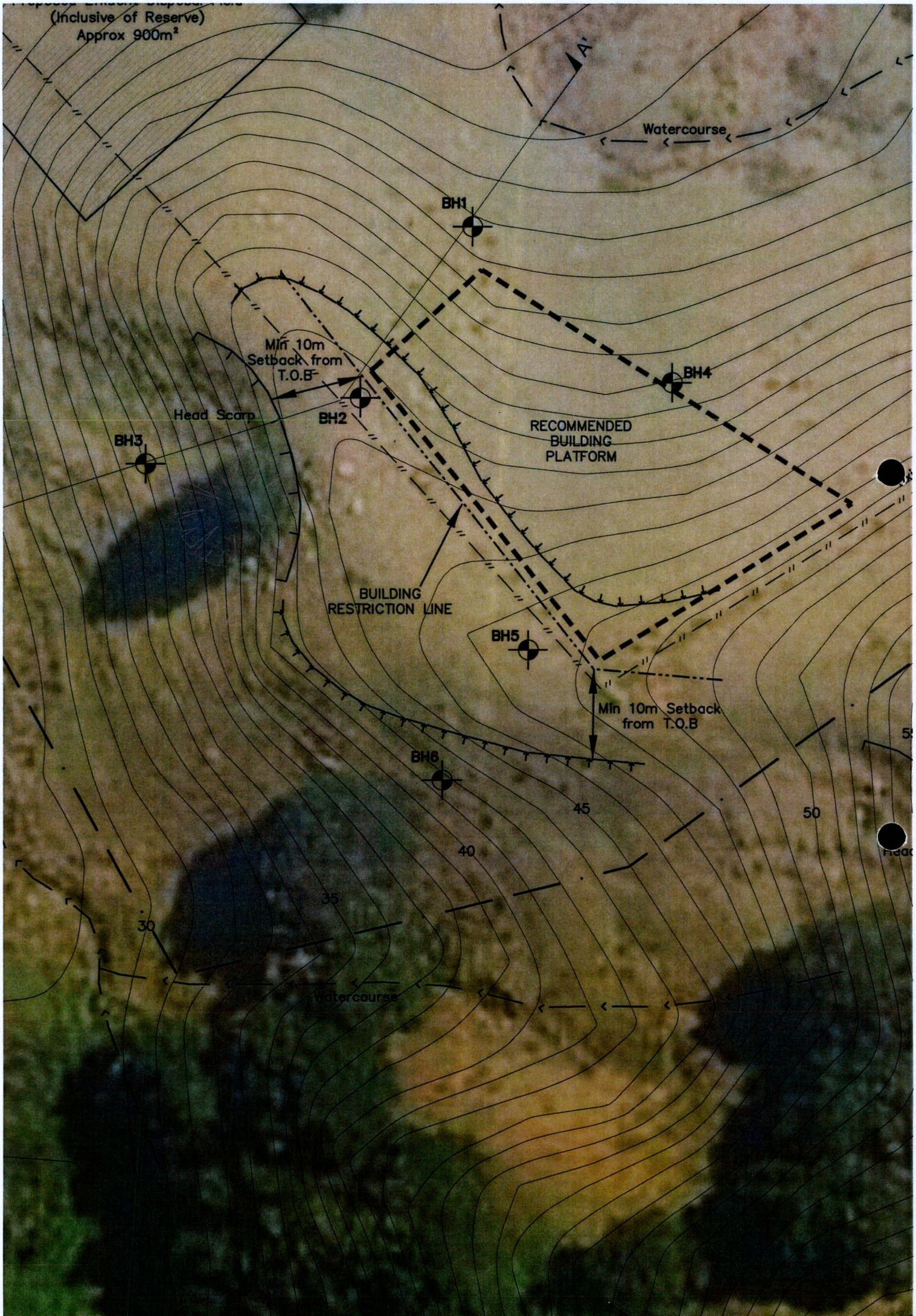
Blows / 100 mm	Depth (m)	Interpret. / Comments
2	4.9	
2		
1	5.1	
3		
4	5.3	
4		
5	5.5	
5		
5	5.7	
5		
8	5.9	
8		
10	6.1	
9		
9	6.3	
10		
11	6.5	
13		
11	6.7	
14		
12	6.9	

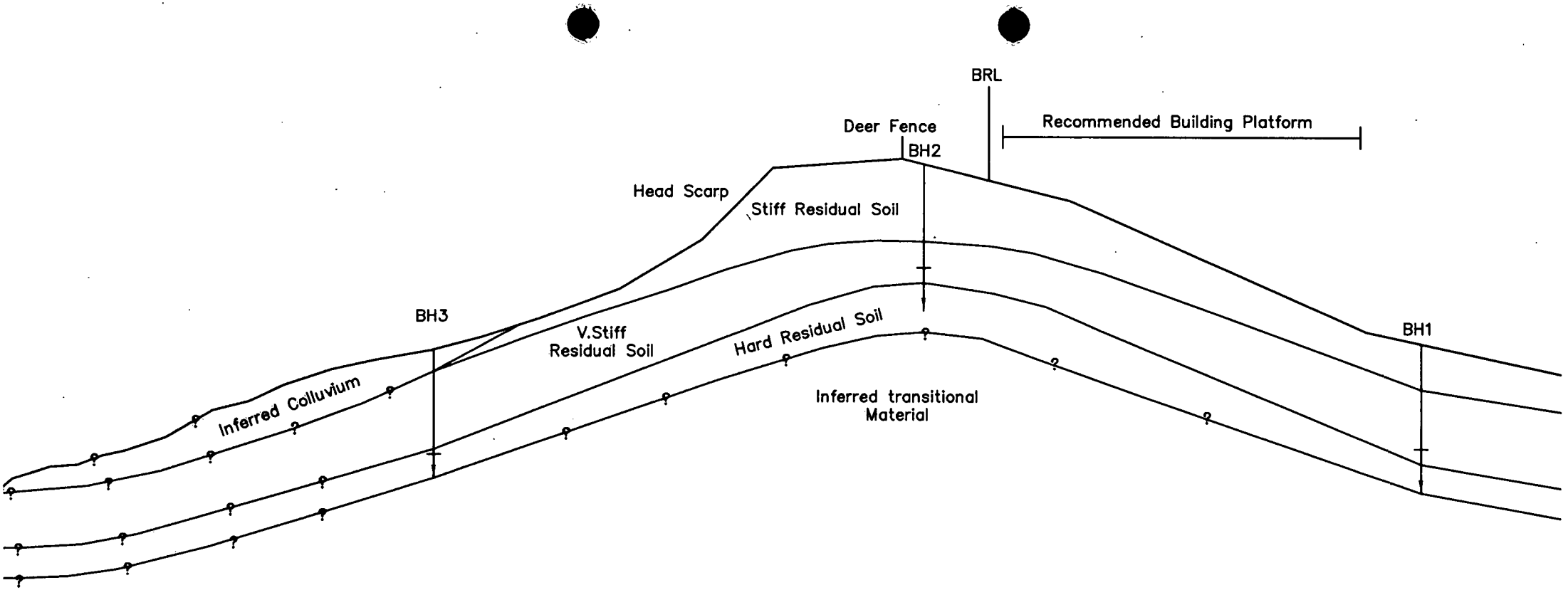
BH 6

Blows / 100 mm	Depth (m)	Interpret. / Comments
13	3.4	
7		
7	3.6	
8		
10	3.8	
5		
8	4.0	
8		
6	4.2	
11		
11	4.4	
15		
15	4.6	
20		
20+	4.8	

NOTES:

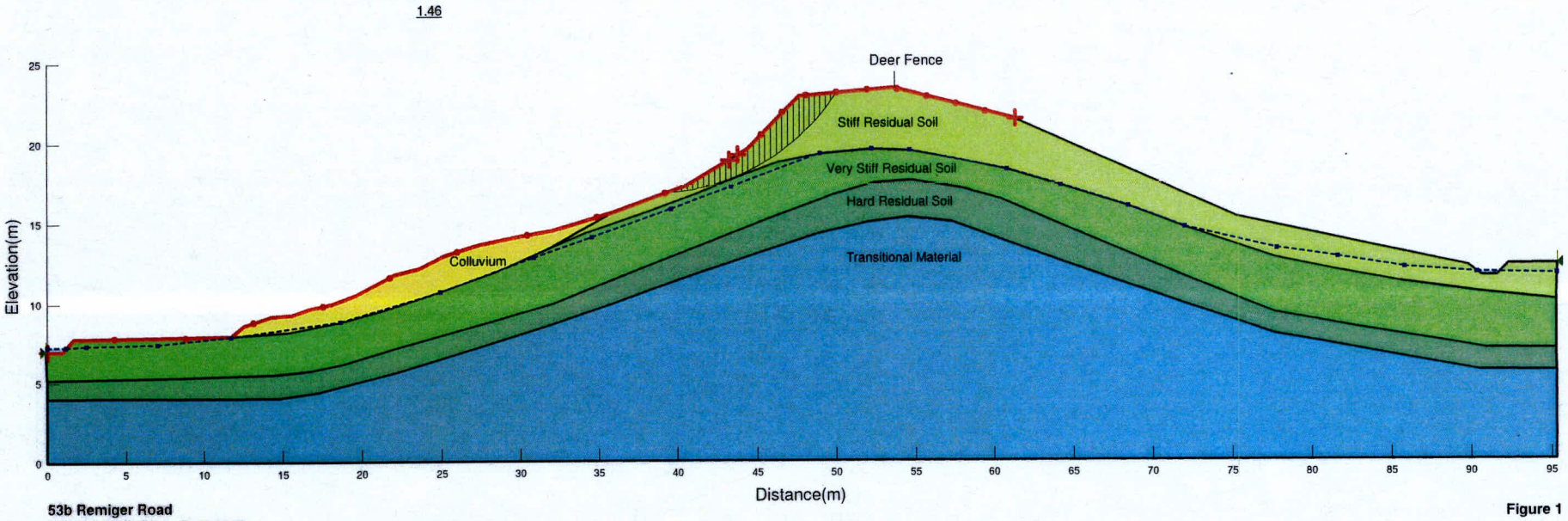
OPERATOR: Jwil/DB
DATE:31/8/2017


(Inclusive of Reserve)
Approx 900m²
A
A'
Watercourse
BH1
Min 10m
Setback from
T.O.B
BH2
BH3
BH4
Head Scarp
RECOMMENDED
BUILDING
PLATFORM
BUILDING
RESTRICTION LINE
BH5
Min 10m Setback
from T.O.B
BH6
45
50
40
35
30
Watercourse



d Level

Level



Name: Colluvium
Unit Weight: 17.5 kN/m³
Cohesion': 3 kPa
Phi': 28 °

Name: Stiff Residual Soil
Unit Weight: 17.5 kN/m³
Cohesion': 5 kPa
Phi': 30 °

Name: Very Stiff Residual Soil
Unit Weight: 18 kN/m³
Cohesion': 7 kPa
Phi': 32 °

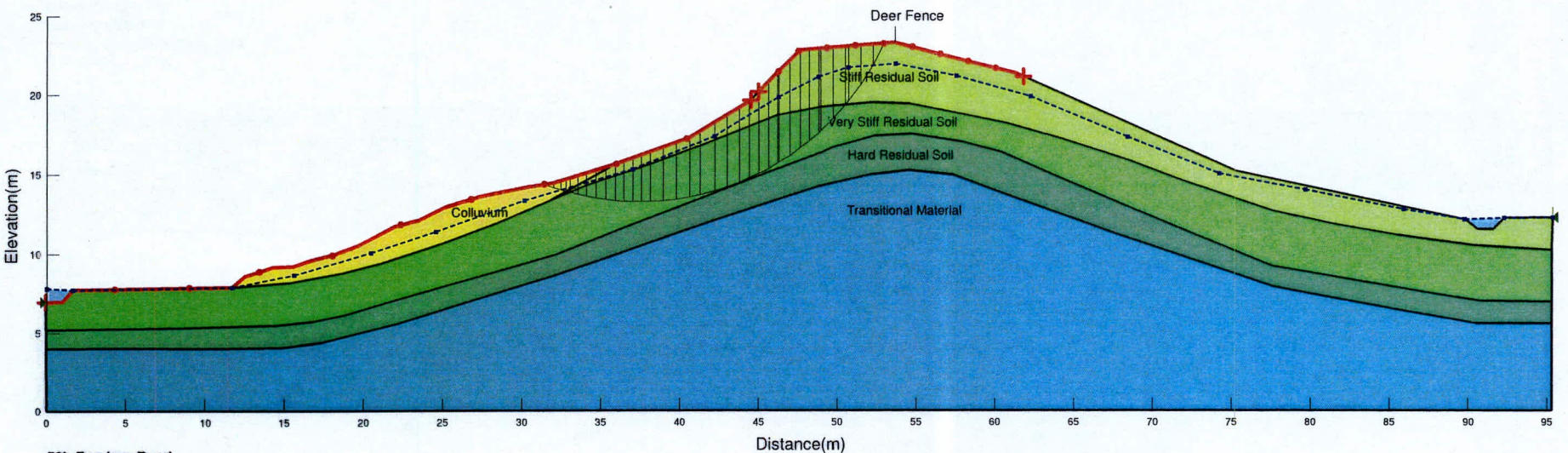
Name: Hard Residual Soil
Unit Weight: 18 kN/m³
Cohesion': 10 kPa
Phi': 34 °

Name: Transitional Material
Unit Weight: 18 kN/m³
Cohesion': 15 kPa
Phi': 36 °

53b Remiger Road
Job No: 8447 Date: Sept 2017
Method: Spencer
Cross Section A-A'
Moderately Convervative Groundwater Table

Figure 1

1.24



Name: Colluvium
Unit Weight: 17.5 kN/m³
Cohesion': 3 kPa
Phi': 28 °

Name: Stiff Residual Soil
Unit Weight: 17.5 kN/m³
Cohesion': 5 kPa
Phi': 30 °

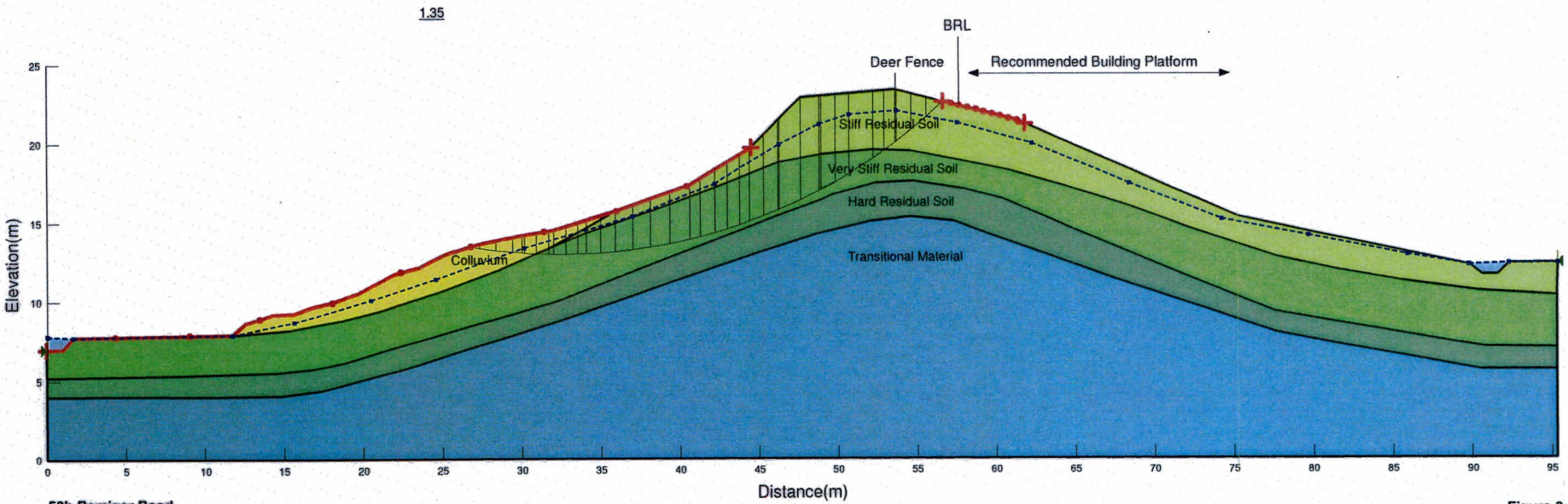
Name: Very Stiff Residual Soil
Unit Weight: 18 kN/m³
Cohesion': 7 kPa
Phi': 32 °

Name: Hard Residual Soil
Unit Weight: 18 kN/m³
Cohesion': 10 kPa
Phi': 34 °

Name: Transitional Material
Unit Weight: 18 kN/m³
Cohesion': 15 kPa
Phi': 36 °

53b Remiger Road
Job No: 8447 Date: Sept 2017
Method: Spencer
Cross Section A-A'
Worst Credible Groundwater Table

Figure 2



Name: Colluvium
Unit Weight: 17.5 kN/m³
Cohesion': 3 kPa
Phi': 28 °

Name: Stiff Residual Soil
Unit Weight: 17.5 kN/m³
Cohesion': 5 kPa
Phi': 30 °

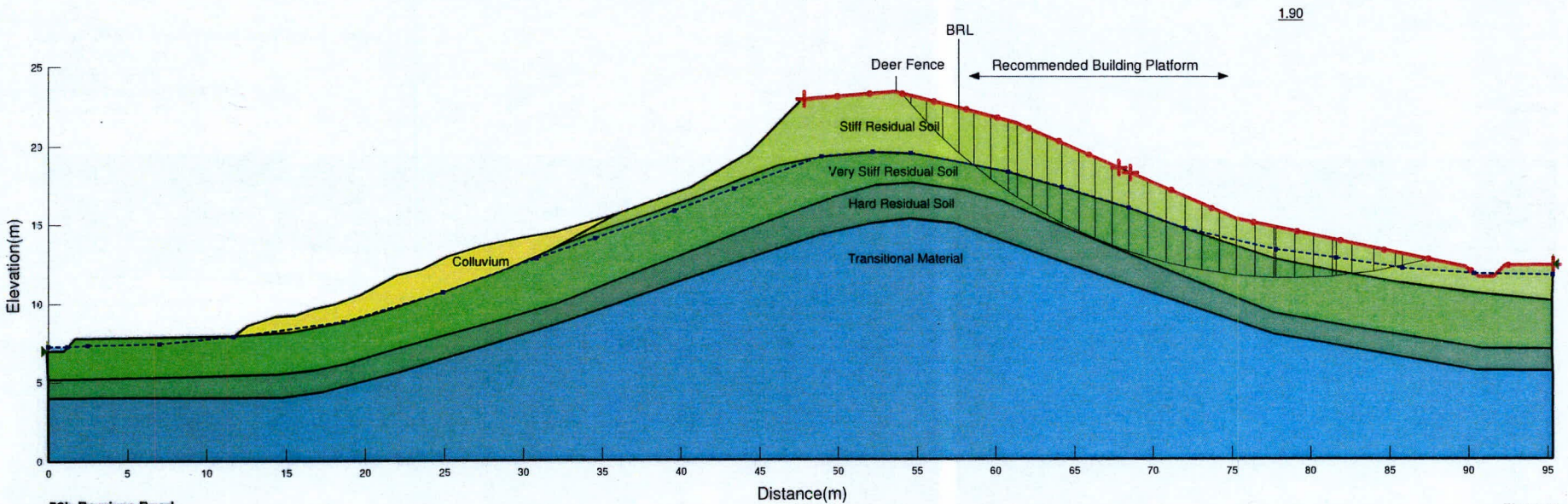
Name: Very Stiff Residual Soil
Unit Weight: 18 kN/m³
Cohesion': 7 kPa
Phi': 32 °

Name: Hard Residual Soil
Unit Weight: 18 kN/m³
Cohesion': 10 kPa
Phi': 34 °

Name: Transitional Material
Unit Weight: 18 kN/m³
Cohesion': 15 kPa
Phi': 36 °

53b Remiger Road
Job No: 8447 Date: Sept 2017
Method: Spencer
Cross Section A-A'
Worst Credible Groundwater Table (BRL)

Figure 3



Name: Colluvium
Unit Weight: 17.5 kN/m³
Cohesion': 3 kPa
Phi': 28 °

Name: Stiff Residual Soil
Unit Weight: 17.5 kN/m³
Cohesion': 5 kPa
Phi': 30 °

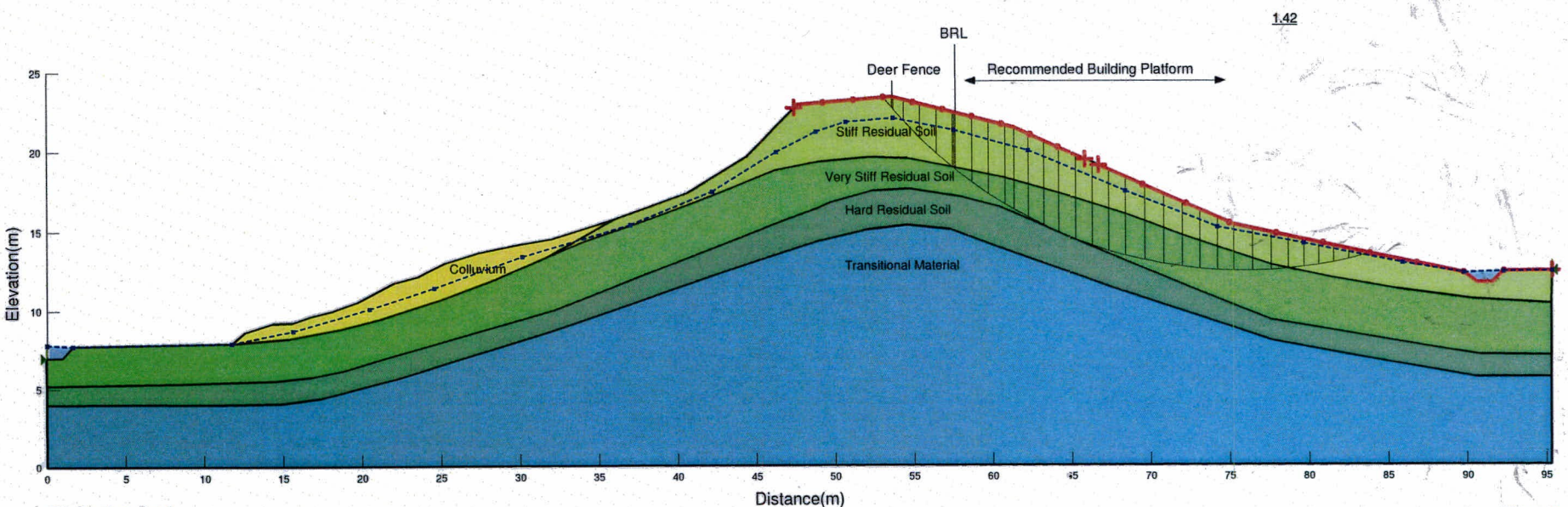
Name: Very Stiff Residual Soil
Unit Weight: 18 kN/m³
Cohesion': 7 kPa
Phi': 32 °

Name: Hard Residual Soil
Unit Weight: 18 kN/m³
Cohesion': 10 kPa
Phi': 34 °

Name: Transitional Material
Unit Weight: 18 kN/m³
Cohesion': 15 kPa
Phi': 36 °

53b Remiger Road
Job No: 8447 Date: Sept 2017
Method: Spencer
Cross Section A-A'
Moderately Convervative Groundwater Table (North Profile)

Figure 4



Name: Colluvium
Unit Weight: 17.5 kN/m³
Cohesion': 3 kPa
Phi': 28 °

Name: Stiff Residual Soil
Unit Weight: 17.5 kN/m³
Cohesion': 5 kPa
Phi': 30 °

Name: Very Stiff Residual Soil
Unit Weight: 18 kN/m³
Cohesion': 7 kPa
Phi': 32 °

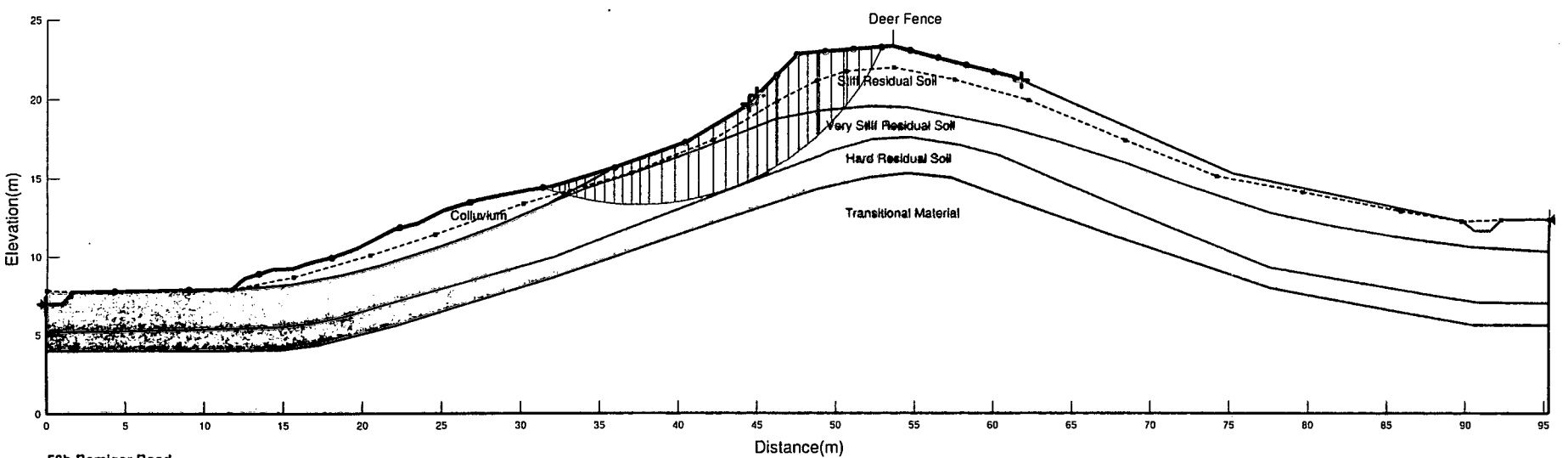
Name: Hard Residual Soil
Unit Weight: 18 kN/m³
Cohesion': 10 kPa
Phi': 34 °

Name: Transitional Material
Unit Weight: 18 kN/m³
Cohesion': 15 kPa
Phi': 36 °

53b Remiger Road
Job No: 8447 Date: Sept 2017
Method: Spencer
Cross Section A-A'
Worst Credible Groundwater Table (North Profile)

Figure 5

1.24



53b Remiger Road
Job No: 8447 Date: Sept 2017
Method: Spencer
Cross Section A-A'
Worst Credible Groundwater Table

Name: Colluvium
Unit Weight: 17.5 kN/m³
Cohesion': 3 kPa
Phi': 28 °

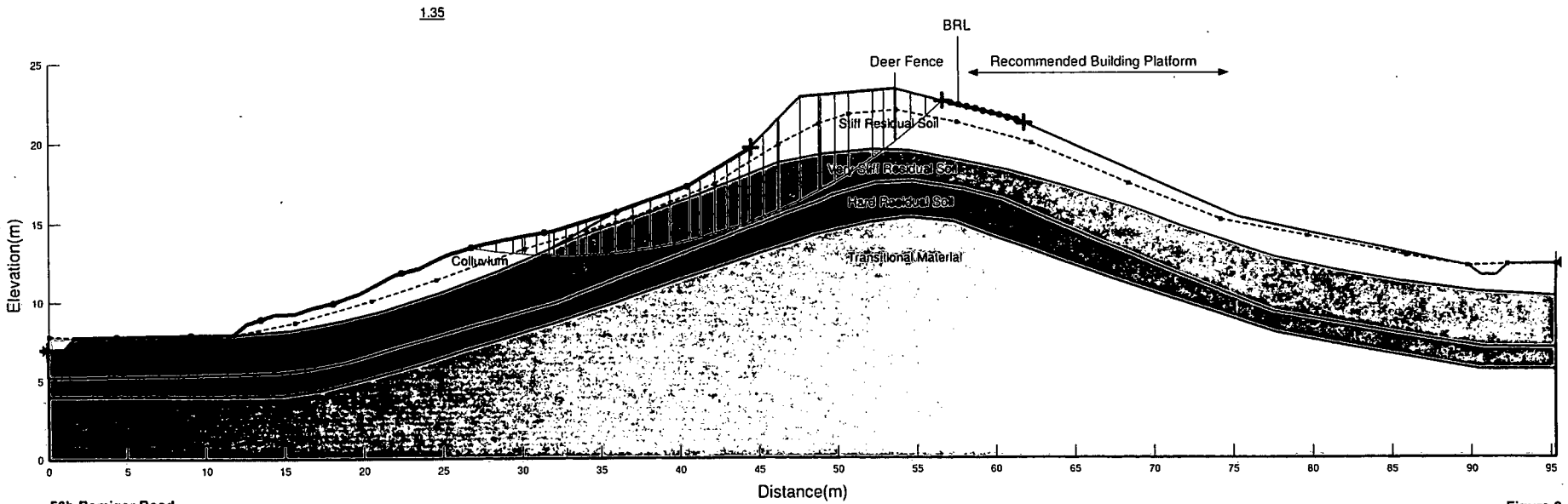
Name: Stiff Residual Soil
Unit Weight: 17.5 kN/m³
Cohesion': 5 kPa
Phi': 30 °

Name: Very Stiff Residual Soil
Unit Weight: 18 kN/m³
Cohesion': 7 kPa
Phi': 32 °

Name: Hard Residual Soil
Unit Weight: 18 kN/m³
Cohesion': 10 kPa
Phi': 34 °

Name: Transitional Material
Unit Weight: 18 kN/m³
Cohesion': 15 kPa
Phi': 36 °

Figure 2



Name: Colluvium
Unit Weight: 17.5 kN/m³
Cohesion': 3 kPa
Phi': 28 °

Name: Stiff Residual Soil
Unit Weight: 17.5 kN/m³
Cohesion': 5 kPa
Phi': 30 °

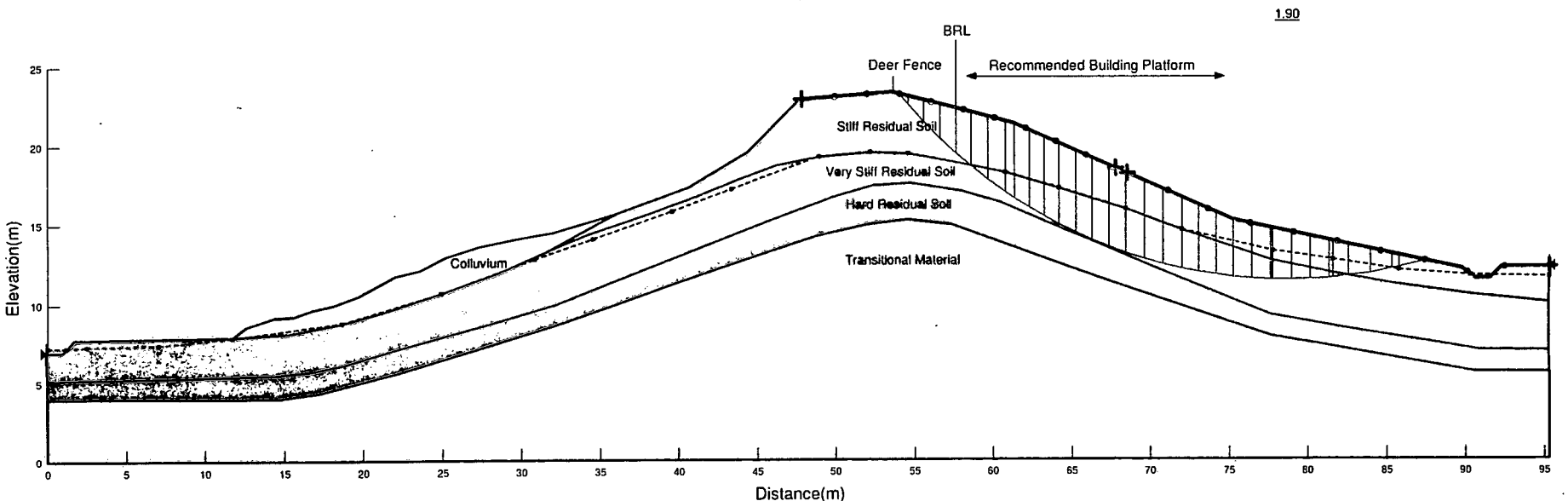
Name: Very Stiff Residual Soil
Unit Weight: 18 kN/m³
Cohesion': 7 kPa
Phi': 32 °

Name: Hard Residual Soil
Unit Weight: 18 kN/m³
Cohesion': 10 kPa
Phi': 34 °

Name: Transitional Material
Unit Weight: 18 kN/m³
Cohesion': 15 kPa
Phi': 36 °

53b Remiger Road
Job No: 8447 Date: Sept 2017
Method: Spencer
Cross Section A-A'
Worst Credible Groundwater Table (BRL)

Figure 3



Name: Colluvium
Unit Weight: 17.5 kN/m³
Cohesion': 3 kPa
Phi': 28 °

Name: Stiff Residual Soil
Unit Weight: 17.5 kN/m³
Cohesion': 5 kPa
Phi': 30 °

Name: Very Stiff Residual Soil
Unit Weight: 18 kN/m³
Cohesion': 7 kPa
Phi': 32 °

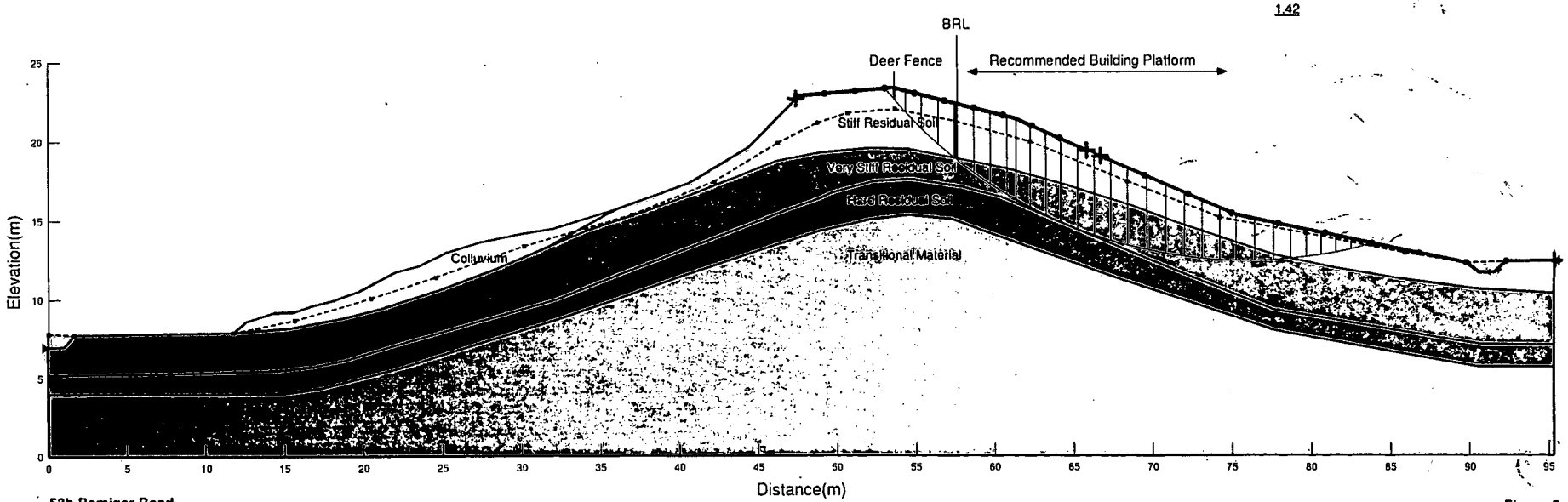
Name: Hard Residual Soil
Unit Weight: 18 kN/m³
Cohesion': 10 kPa
Phi': 34 °

Name: Transitional Material
Unit Weight: 18 kN/m³
Cohesion': 15 kPa
Phi': 36 °

53b Remiger Road
Job No: 8447 Date: Sept 2017
Method: Spencer
Cross Section A-A'
Moderately Convervative Groundwater Table (North Profile)

Figure 4

1.42



Name: Colluvium
Unit Weight: 17.5 kN/m³
Cohesion': 3 kPa
Phi': 28 °

Name: Stiff Residual Soil
Unit Weight: 17.5 kN/m³
Cohesion': 5 kPa
Phi': 30 °

Name: Very Stiff Residual Soil
Unit Weight: 18 kN/m³
Cohesion': 7 kPa
Phi': 32 °

Name: Hard Residual Soil
Unit Weight: 18 kN/m³
Cohesion': 10 kPa
Phi': 34 °

Name: Transitional Material
Unit Weight: 18 kN/m³
Cohesion': 15 kPa
Phi': 36 °

53b Remiger Road
Job No: 8447 Date: Sept 2017
Method: Spencer
Cross Section A-A'
Worst Credible Groundwater Table (North Profile)

Figure 5